Session Notes — A2A Protocol Deep Dive

Oct 25, 2026 · @kumar

A2A (Agent-to-Agent) is Google's 2025 protocol for standardizing communication between agents running on different servers.

MCP vs A2A

	MCP	A2A
Introduced by	Anthropic	Google
Year	2024	2025
Purpose	Provide context, tools, and resources *to* a model	Standardize communication *between* agents
Key components	Client + Server	Client + Server
What it handles	Tool calls, prompts, external resources	Cross-agent requests over a network

When A2A Is and Isn't Needed

The rule: if both agents are on the same machine in the same application, A2A is not needed — a direct function call (`agent.invoke()`) works. A2A is needed when agents are on different servers.

Why `invoke()` fails across servers: `invoke()` is a function call inside the same Python process, reaching into shared memory. When Agent B is on a different machine, there is no shared memory or process — you cannot call a function that lives on another server. You need a network-level request to bridge them.

Situation	A2A needed?
Both agents, same app, same server	No — direct handoff works
Two apps on the same server	Optional — A2A can be used

Situation	A2A needed?
Agents on different servers/machines	Yes

Without A2A: you can use a plain REST API (`requests.post(server_url, …)`). It works, but every agent may have a different API structure. With 100 companies' agents, you'd need custom integration code for each.

With A2A: every compliant agent exposes a standard interface. One client can talk to any server without custom code.

Protocol Bindings

A **protocol binding** is the specific mechanism chosen to carry A2A communication rules over the network. You pick **one** per application — not all three.

Binding	Full name	Format	Speed	Best for
JSON-RPC	JavaScript Object Notation Remote Procedure Call	Text (JSON)	Moderate	Most common; readable; default in the A2A package
gRPC	Remote Procedure Call (Google)	Binary (Protocol Buffers)	Fastest	High-frequency, performance-critical systems
HTTP + JSON	Hypertext Transfer Protocol + JSON	Text (JSON over REST)	Moderate	Simplest; familiar to most developers

Key difference — JSON-RPC vs HTTP + JSON: both use JSON, but:

- JSON-RPC: the method name is specified *inside* the JSON body
- HTTP + JSON: the operation is identified by the HTTP method (GET/POST) and the URL endpoint

The demo from the Oct 24 session used **JSON-RPC**.

Important A2A Terms

Example used in class: user asks travel agent → "Find flights from Bangalore to Dubai". Travel agent calls the flight agent.

Term	Definition	In the example
A2A Client	The agent that initiates and sends the request	Travel agent
A2A Server	The agent that receives and processes the request	Flight agent
Agent Card	A digital profile of the agent — name, description, capabilities	Flight agent's identity card
Agent Skill	What the agent is capable of doing	"Search and return flight data"
Agent Discovery	The process of finding available agents on the network	—
Message	The actual content exchanged between agents	"Find flights from Bangalore to Dubai"
Task	A trackable unit of work assigned to an agent	The flight search request
Task ID	Unique identifier for a specific task	—
Task Status	Current execution state of the task	pending / running / complete
Artifact	The output or deliverable generated during task execution	The list of available flights returned
Context ID	Groups related conversations and tasks under one identifier	Links follow-up queries to the same session

Next Topic

Evaluation — covered in the session after this one.